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Low-Rank Scale-Invariant Tensor Product Smooths for Generalized Additive Mixed Models

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Summary. A general method for constructing low-rank tensor product smooths for use as components of generalized additive models or generalized additive mixed models is presented. A penalized regression approach is adopted in which tensor product smooths of several variables are constructed from smooths of each variable separately, these "marginal" smooths being represented using a low-rank basis with an associated quadratic wiggliness penalty. The smooths offer several advantages: (i) they have one wiggliness penalty per covariate and are hence invariant to linear rescaling of covariates, making them useful when there is no "natural" way to scale covariates relative to each other; (ii) they have a useful tuneable range of smoothness, unlike single-penalty tensor product smooths that are scale invariant; (iii) the relatively low rank of the smooths means that they are computationally efficient; (iv) the penalties on the smooths are easily interpretable in terms of function shape; (v) the smooths can be generated completely automatically from any marginal smoothing bases and associated quadratic penalties, giving the modeler considerable flexibility to choose the basis penalty combination most appropriate to each modeling task; and (vi) the smooths can easily be written as components of a standard linear or generalized linear mixed model, allowing them to be used as components of the rich family of such models implemented in standard software, and to take advantage of the efficient and stable computational methods that have been developed for such models. A small simulation study shows that the methods can compare favorably with recently developed smoothing spline ANOVA methods.

Key words: Computationally efficient; Generalized additive mixed model (GAMM); Mixed effect variable coefficient model; Multiple penalties; Penalized regression; Scale invariant; Smooth interaction; Smoothing penalty; Spline; SS-ANOVA; Tensor product smooth.

1. Introduction

An additive mixed model (special case of a generalized additive mixed model [GAMM]; Lin and Zhang, 1999; Fahrmeir and Lang, 2001) has a structure something like

$$y_i = \mathbf{X}_i\boldsymbol{\theta} + w_{1i}f_1(x_{1i}) + w_{2i}f_2(x_{2i}, x_{3i}) + \cdots + \mathbf{Z}_i\mathbf{b} + \epsilon_i, \quad (1)$$

where y_i is a univariate response; $\boldsymbol{\theta}$ is a vector of fixed parameters; $\mathbf{X}_i$ is a row of a fixed effects model matrix; the w_{ji}'s are covariates, dummy variables, or often simply 1 (they are used in "variable coefficient models"; Hastie and Tibshirani, 1993); the f_j's are smooth functions of covariates x_k; $\mathbf{Z}_i$ is a row of a random effects model matrix; $\mathbf{b} \sim N(0, \boldsymbol{\psi})$ is a vector of random effects coefficients with unknown positive definite covariance matrix $\boldsymbol{\psi}$; and $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \boldsymbol{\Lambda})$ is a residual error vector, with ith element ϵ_i and covariance matrix $\boldsymbol{\Lambda}$, which is usually assumed to have some simple pattern. *Generalized* additive mixed models replace the normal residuals assumption with an assumption that $y_i \,|\, \mathbf{b}$ has some exponential family distribution and $E(y_i \,|\, \mathbf{b})$ is some monotonic function of the right-hand side of (1), excluding the ϵ_i term. These models are closely related to the geoadditive models of Kammann and Wand (2003), and Ruppert, Wand, and Carroll (2003) discuss a number of examples of models of this type. Additive mixed models also bear some relation to the models for designed experiments discussed, for example, by Verbyla et al. (1999) and implemented, for example, by Ball (2003). Generalized additive models (GAMs; Hastie and Tibshirani, 1990; see also Wahba, 1990) are a special case of GAMMs, which have no $\mathbf{Z}_i\mathbf{b}$ term.

GAMMs have an advantage over GAMs in that the more complex stochastic structure allows treatment of autocorrelation and repeated measures situations. The way in which smooths are actually incorporated into GAMMs varies. Lin and Zhang (1999) used cubic smoothing splines to represent the univariate smooths that they considered, while Wang (1998) represented a full smoothing spline analysis of variance (SS-ANOVA) model (see, e.g., Gu, 2002) as a normal linear mixed model. But other authors have tended to opt for the more computationally parsimonious penalized regression splines, either P-splines (Eilers and Marx, 1996) estimated using Markov chain Monte Carlo (MCMC; Fahrmeir and Lang, 2001) or some variant on the thin plate spline basis or truncated power bases with estimation by restricted maximum likelihood estimation (REML; Kammann and Wand, 2003; Ruppert et al., 2003).

Three approaches to representing smooths of more than one variable in GAMMs have been suggested. Either

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low-rank approximations to thin plate splines have been employed (Kammann and Wand, 2003; Ruppert et al., 2003) or tensor product P-splines have been suggested, with the single penalty given by the Kronecker product of the penalties associated with the marginal bases from which the smoothing basis is constructed (Fahrmeir and Lang, 2001). Finally, for smooths of two predictors in a fully Bayesian setting, and recognizing the undersmoothing that results from single Kronecker product penalties, Lang and Brezler (2004) suggested employing tensor products of equally spaced B-spline basis functions in conjunction with spatially symmetric priors on the B-spline coefficients based on neighboring coefficients. Lang and Brezler also generalized this to allow the degree of smoothing to vary over space: these smooths perform well but are not invariant to rescaling of the covariates.

By contrast, in non-GAMM settings the full tensor product smoothing splines of Wahba (1990) and Gu (2002) have a number of smoothing penalties associated with each component of an "ANOVA decomposition" of a smooth function, in such a way that the resulting smooths are invariant to covariate rescaling, but this approach is not easy to integrate with conventional approaches to (generalized) linear mixed modeling. Eilers and Marx (2003) have also used tensor products of B-splines to represent two-dimensional surfaces, with separate difference penalties applied to the coefficients of the B-splines along the two covariate axes. When it is not appropriate to assume isotropy of a smooth of several variables then the invariance of such tensor product smooths is an important property.

The aims of this article are (i) to produce a general framework for constructing "scale-invariant" tensor product smooths from low-rank penalized regression smoothers consisting of any set of basis functions and associated quadratic "wiggliness" penalties, and (ii) to produce a general method for incorporating such smooths into standard (generalized) linear mixed models. This should substantially increase the flexibility both of smooth models, by allowing direct access to existing mixed model components and methods, and of mixed modeling, by giving a straightforward method of modeling smooth interactions, without the need for a complete change in modeling framework. The main novelty of the work lies in the general (and "natural") method of producing wiggliness penalties for a tensor product smooth from wiggliness penalties on the "marginal bases" used to generate it. The method for expressing such smooths as mixed model components is also novel: the general idea of expressing smooths as mixed model components is well established, but in most cases relies on the smoothing basis separating naturally into penalized and unpenalized components. In the current context no such separation can be assumed, so that a general reparameterization strategy is required in order for the proposed smooths to be treated as mixed model components within a conventional mixed model framework. Finally, note the important point that low-rank smoothers are considered in order to gain computational efficiency (or sometimes even feasibility) relative to, for example, full spline models. Full spline models require one model coefficient per datum, and hence have computational cost scaling as the cube of the number of data: the methods discussed here can have far fewer coefficients, and consequently lower cost.

2. Low-Rank Tensor Product Smooths

This section develops general methods for producing penalized regression smoothers, which are functions of several variables. The smooths have the following properties: (i) they are invariant to linear rescaling of their covariates (while having a useful adjustable smoothness range); (ii) they are of relatively low rank, so that their computation is efficient and feasible for any size of data set; (iii) they can be readily incorporated into existing penalized regression and mixed modeling frameworks, so that they add modeling functionality to these frameworks, rather than requiring the development of separate theory for fitting and inference; (iv) they allow the modeler considerable flexibility to easily construct smoothers designed for particular applied modeling tasks; and (v) the penalties are fairly easy to interpret.

The basic approach of this section is to start from smooths of single covariates, represented using any basis with associated quadratic penalty measuring "wiggliness" of the smooth. From these "marginal smooths" a "tensor product" construction is used to build up smooths of several variables. The basis construction is straightforward (and not novel): the key innovation is a method for obtaining simple wiggliness penalties for the tensor product smooth, which are interpretable in terms of function shape and are induced naturally and automatically from the penalties of the marginal smooths.

The methods developed here can be used to construct smooth functions of *any* number of covariates, but the simplest introduction is via the construction of a smooth function of three covariates, x, z, and v, the generalization then being trivial. The process starts by assuming that we have available low-rank bases for representing smooth functions f_x, f_z, and f_v of each of the covariates. That is, we can write

$$f_x(x) = \sum_{i=1}^{I} \alpha_i a_i(x), \quad f_z(z) = \sum_{j=1}^{J} \delta_j d_j(z), \quad \text{and}$$

$$f_v(v) = \sum_{k=1}^{K} \beta_k b_k(v),$$

where the α_i, δ_j, and β_k are parameters and the $a_i(x)$, $d_j(z)$, and $b_k(v)$ are known basis functions.

Now consider how the smooth function of x, f_x, could be converted into a smooth function of x and z. What is required is for f_x to vary smoothly with z, and this can be achieved by allowing its parameters, α_i, to vary smoothly with z. Using the basis already available for representing smooth functions of z we could write

$$\alpha_i(z) = \sum_{j=1}^{J} \delta_{ij}\, d_j(z),$$

which immediately gives

$$f_{xz}(x, z) = \sum_{i=1}^{I} \sum_{j=1}^{J} \delta_{ij} d_j(z) a_i(x).$$

Continuing in the same way, we could now create a smooth function of x, z, and v by allowing f_{xz} to vary smoothly with v. Again, the obvious way to do this is to let the parameters of

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f_{xz} vary smoothly with v, and following the same reasoning as before we get

$$f_{xzv}(x,z,v) = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} \beta_{ijk} b_k(v)\, d_j(z) a_i(x).$$

For any particular set of observations of x, z, and v a model matrix, $\mathbf{X}$, can be produced, which maps the parameters β_{ijk} (suitably arranged into a vector $\boldsymbol{\beta}$) to the evaluated function at these values. If $\mathbf{X}_x$, $\mathbf{X}_z$, and $\mathbf{X}_v$ are model matrices for the individual marginal smooths, and $\otimes$ is the usual Kronecker product, then it is easy to show that, given appropriate ordering of $\boldsymbol{\beta}$, the ith row of $\mathbf{X}$ is simply

$$\mathbf{X}_i = \mathbf{X}_{xi} \otimes \mathbf{X}_{zi} \otimes \mathbf{X}_{vi}.$$

Clearly, (i) this construction can be continued for as many covariates as are required; (ii) the result is independent of the order in which we treat the covariates; and (iii) the covariates can themselves be vector covariates.

Having derived a "tensor product" basis for representing smooth functions, it is also necessary to have some way of measuring function "wiggliness," if the basis is to be useful for representing smooth functions in a penalized regression or mixed modeling context. Again it is possible to start from wiggliness measures associated with the marginal smooth functions, and again the three-covariate case provides sufficient illustration. Suppose then, that each marginal smooth has an associated functional that measures function wiggliness and can be expressed as a quadratic form in the marginal parameters. That is

$$J_x(f_x) = \boldsymbol{\alpha}^{\mathrm{T}}\mathbf{S}_x\boldsymbol{\alpha}, \quad J_z(f_z) = \boldsymbol{\delta}^{\mathrm{T}}\mathbf{S}_z\boldsymbol{\delta}, \quad J_v(f_v) = ß^{\mathrm{T}}\mathbf{S}_v ß.$$

The $\mathbf{S}_\bullet$ matrices contain known coefficients, and $\boldsymbol{\alpha}$, $\boldsymbol{\delta}$, and ß are vectors of coefficients of the marginal smooths. An example of a penalty functional is the cubic spline penalty, $J_x(f_x) = \int (\partial^2 f_x/\partial x^2)^2\, dx$. Now let $f_{x|zv}(x)$ be $f_{xvz}(x,\ z,\ v)$ considered as a function of x only, with z and v held constant, and define $f_{z|xv}(z)$ and $f_{v|xz}(v)$ similarly. A natural way of measuring wiggliness of f_{xzv} is to use

$$J(f_{xzv}) = \lambda_x \int_{z,v} J_x(f_{x|zv})\, dz\, dv + \lambda_z \int_{x,v} J_z(f_{z|xv})\, dx\, dv + \lambda_v \int_{x,z} J_v(f_{v|xz})\, dx\, dz,$$

where the $\lambda_\bullet$ are smoothing parameters controlling the trade-off between wiggliness in different directions, and allowing the penalty to be invariant to the relative scaling of the covariates. The mass of subscripts in the preceding expression can obscure the fact that what is being done here is a rather straightforward and natural way of generalizing from marginal penalties: Figure 1a attempts to clarify this in the context of

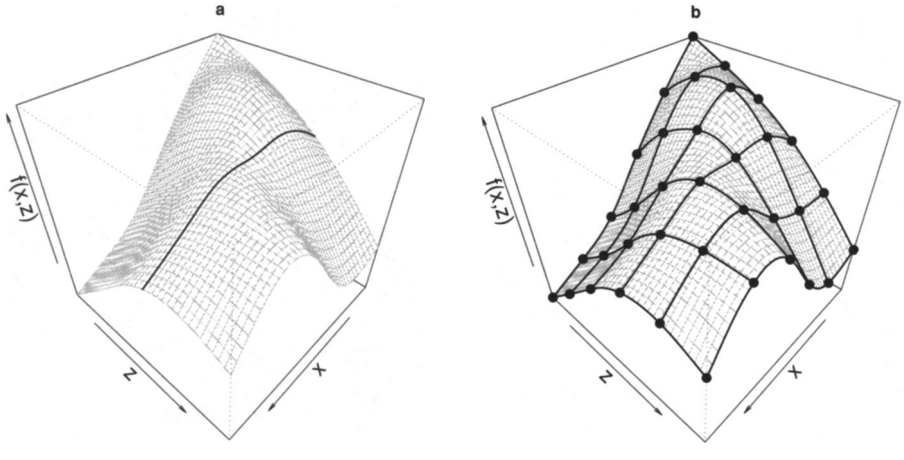

Figure 1. Two-covariate illustration of the principles underlying the proposed tensor product smooth penalties. (a) The gray surface illustrates a tensor product smooth of two covariates x and z. Consider how to measure wiggliness of this surface with respect to x. It is always possible to hold z constant and consider variation of the surface with respect to x alone, that is, to consider the variation of the function of x, $f_{x|z}(x)$. An example $f_{x|z}(x)$ is illustrated by the thick curve on the surface. Now, if the marginal basis for x has an associated penalty for measuring wiggliness with respect to x then this can obviously be used to measure wiggliness of any particular $f_{x|z}(x)$ with respect to x. A natural way to measure wiggliness of the whole surface with respect to x is simply to find the average, over all z, of the wiggliness of all curves like $f_{x|z}(x)$, and this is the basis of the penalty construction used in this article. Obviously, a symmetric construction holds for the z penalty. (b) Representation of a tensor product spline using the reparameterization and penalty approximation suggested in the text. The reparameterization means that, however the underlying marginal bases are constructed, the tensor product smooth is represented in terms of function values on a regular mesh (illustrated by • on the figure). The approximation of the integral in the penalty means that function wiggliness with respect to x is measured by the average wiggliness with respect to x of the thick curves parallel to the x-axis that are shown on the surface of the tensor product smooth (each curve's wiggliness being measured by the marginal measure of wiggliness with respect to x). The example was constructed from a rank 6 marginal smooth of z, so wiggliness with respect to x is measured from the mean wiggliness of six curves, in this case. Symmetric definitions apply to wiggliness with respect to z.

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a two-covariate smooth. As a three-covariate example, if cubic spline penalties were used as the marginal penalties, then

$$J(f) = \int_{x,z,v} \lambda_x \left(\frac{\partial^2 f}{\partial x^2}\right)^2 + \lambda_z \left(\frac{\partial^2 f}{\partial z^2}\right)^2 + \lambda_v \left(\frac{\partial^2 f}{\partial v^2}\right)^2 dx\, dz\, dv.$$

Hence, if the marginal penalties are easily interpretable in terms of function shape, then so is the induced penalty. Numerical evaluation of the integrals in J is straightforward. As an example consider the penalty in the x direction. The function $f_{x|zv}(x)$ can be written as

$$f_{x|zv}(x) = \sum_{i=1}^{I} \alpha_i(z,v) a_i(x)$$

and it is always possible to find the matrix of coefficients $\mathbf{M}_{z,v}$ such that $\boldsymbol{\alpha}(z, v) = \mathbf{M}_{zv}\boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is the vector of β_{ijk} arranged in some appropriate order. Hence

$$J_x(f_{x|zv}) = \boldsymbol{\alpha}(z,v)^{\mathrm{T}}\mathbf{S}_x\boldsymbol{\alpha}(z,v) = \boldsymbol{\beta}^{\mathrm{T}}\mathbf{M}_{zv}^{\mathrm{T}}\mathbf{S}_x\mathbf{M}_{zv}\boldsymbol{\beta}$$

and so

$$\int_{z,v} J_x(f_{x|zv})\, dz\, dv = \boldsymbol{\beta}^{\mathrm{T}} \int_{z,v} \mathbf{M}_{zv}^{\mathrm{T}}\mathbf{S}_x\mathbf{M}_{zv}\, dz\, dv\boldsymbol{\beta}.$$

The last integral can be performed numerically, and it is clear that the same approach can be applied to all components of the penalty. However, a simple reparameterization can be used to provide an approximation to the terms in the penalty, which performs well in practice and avoids the need for explicit numerical integration.

To see how the approach works, consider the marginal smooth f_x. Let $\{x_i^* : i = 1, \ldots, I\}$ be a set of values of x spread evenly through the range of the observed x values. In this case we can always reparameterize f_x in terms of new parameters:

$$\alpha_i' = f_x\left(x_i^*\right).$$

Clearly, under this reparameterization $\boldsymbol{\alpha}' = \mathbf{A}\boldsymbol{\alpha}$, where $A_{ij} = a_i(x_j^*)$. Hence the marginal model matrix becomes $\mathbf{X}_x' = \mathbf{X}_x\mathbf{A}^{-1}$ and the penalty coefficient matrix becomes $\mathbf{S}_x' = \mathbf{A}^{-\mathrm{T}}\mathbf{S}_x\mathbf{A}^{-1}$.

Now suppose that the same sort of reparameterization is applied to the marginal smooths f_v and f_z. In this case we have that

$$\int_{z,v} J_x(f_{x|zv})\, dz\, dv \approx h \sum_{jk} J_x\left(f_{x|z_j^* v_k^*}\right),$$

where h is some constant of proportionality related to the spacing of the z_j^*'s and v_k^*'s. Similar expressions hold for the other integrals making up $J(f)$. It is straightforward to show that the summation in the above approximation is

$$J_x^*(f_{xzv}) = \boldsymbol{\beta}^{\mathrm{T}}\tilde{\mathbf{S}}_x\boldsymbol{\beta}, \text{where } \tilde{\mathbf{S}}_x = \mathbf{S}_x' \otimes \mathbf{I}_J \otimes \mathbf{I}_K,$$

and $\mathbf{I}_J$ is the rank J identity matrix. Exactly similar definitions hold for the other components of the penalty so that

$$J_z^*(f_{xzv}) = \boldsymbol{\beta}^{\mathrm{T}}\tilde{\mathbf{S}}_z\boldsymbol{\beta}, \text{where } \tilde{\mathbf{S}}_z = \mathbf{I}_I \otimes \mathbf{S}_z' \otimes \mathbf{I}_K$$

and

$$J_v^*(f_{xzv}) = \boldsymbol{\beta}^{\mathrm{T}}\tilde{\mathbf{S}}_v\boldsymbol{\beta}, \text{where } \tilde{\mathbf{S}}_v = \mathbf{I}_I. \otimes \mathbf{I}_J \otimes \mathbf{S}_v'.$$

Hence

$$J(f_{xzv}) \approx J^*(f_{xzv}) = \lambda_x J_x^*(f_{xzv}) + \lambda_z J_z^*(f_{xzv}) + \lambda_v J_v^*(f_{xzv}),$$

where any constants, h, have been absorbed into the λ_j. Again, this penalty construction clearly generalizes to any number of covariates. Figure 1b provides a visual illustration and explanation of what the reparameterization and penalty approximation actually results in, which is probably more immediately accessible than the mathematical argument alone.

Given its model matrix and penalties, the coefficients and smoothing parameters of a tensor product smooth can be estimated using the multiple penalized regression methods given in Wood (2004), or incorporated into a generalized additive model and estimated in the same way. Alternatively, the smooth can be reparameterized for representation as a component of a mixed model, as shown in Section 3. Whatever the estimation method, the smooths are invariant to rescaling of the covariates, provided only that the marginal smooths are similarly invariant (which is always the case in practice).

2.1 *Nesting and ANOVA Decompositions*

Consider a set of marginal smoothing bases used to construct interaction smooths of various orders. The tensor product construction immediately implies that any interaction of some set of covariates is nested within all higher-order interactions including that set of covariates (subject only to the mild technical restriction that all marginal bases include the constant function in their span). So, for example, the model $y_i = f_x(x_i) + f_z(z_i) + \epsilon_i$ is strictly nested within the model $y_i = f_{zx}(x_i, z_i) + \epsilon_i$ provided that the bases used for f_x and f_z are the marginal bases used for f_{xz}. This immediately opens up the possibility of working with "ANOVA decomposition"-type models:

$$y_i = f_x(x_i) + f_z(z_i) + f_{zx}(x_i, z_i) + \epsilon_i,$$

and

$$y_i = f_x(x_i) + f_z(z_i) + f_v(v_i) + f_{xz}(x_i, z_i) + f_{xv}(x_i, v_i) + f_{zv}(z_i, v_i) + f_{xzv}(x_i, z_i, v_i) + \epsilon_i,$$

etc., because all that is required to do this is to generate model matrices and penalties for each term and then to impose appropriate identifiability constraints so that the whole model is identifiable (obviously higher-order interactions will often be excluded from the model at the outset). For example, if f_{zx} is created from a tensor product of the bases used to represent f_x and f_z, then its basis automatically spans the space of functions representable by the basis for $f_x(x) + f_z(z)$: to make identifiable a model including the terms $f_x(x) + f_z(z) + f_{zx}(x, z)$ therefore requires that we impose linear constraints on the basis/coefficients of f_{zx} so that it becomes independent of the bases for the terms f_x and f_z: calculation and imposition of such constraints is routine. The resultant reparameterization leaves unchanged the interpretation of the penalties on f_{zx}, although the interpretation of f_{xz} itself is modified, of course. When using such decompositions, each component smooth will obviously have

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one smoothing parameter associated with each of its penalties (e.g., the three-covariate decomposition given above would have 12 smoothing parameters).

This notion of an ANOVA decomposition of functions has been pioneered in the smoothing spline literature, and is the subject of a monograph by Gu (2002). Fuller comparisons with existing SS-ANOVA methods are provided in Section 4.2.

3. Tensor Product Smooths as Mixed Model Components

The aim of this section is to show how the tensor product smooths, as given in Section 2, can be included as components of a linear mixed model of the form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\epsilon}, \text{ where } \mathbf{b} \sim N(\mathbf{0}, \boldsymbol{\psi}),\ \boldsymbol{\epsilon} \sim N(\mathbf{0}, \boldsymbol{\Lambda}), \qquad (2)$$

and $\mathbf{y}$ is a response vector; $\mathbf{X}$ and $\mathbf{Z}$ are model matrices; $\boldsymbol{\beta}$ and $\mathbf{b}$ are fixed parameters and random effects, respectively; and $\boldsymbol{\epsilon}$ is a residual error vector. The covariance matrix $\boldsymbol{\psi}$ is usually parameterized in terms of some unknown parameters, while $\boldsymbol{\Lambda}$ is usually either diagonal or has some other simple form induced by a simple correlation model. The idea is to add to the already extensive range of models readily available in this framework by providing straightforward means to include a wide range of tensor product smooths in such models.

The representation of penalized regression smoothers as mixed model components is now well established (see, e.g., Ruppert et al., 2003, for a good introduction); however, most of the literature in this area has relied on the use of smoothing bases that were already naturally separated into some components identifiable as fixed effects, and others identifiable as random effects. The tensor product smooths as given in Section 2 and many useful penalized regression smoothers are not in such convenient form (e.g., P-splines, some pseudo-splines, or even some otherwise convenient representations of cubic splines), so that reparameterization is necessary if they are to be employed in this way. Recently, appropriate general reparameterization approaches for smooths with a single penalty have been presented (Fahrmeir, Kneib, and Lang, 2004 or the Appendix of Wood, 2004), and the purpose of this section is to find an appropriate general reparameterization scheme for use with the tensor product smooths as given in Section 2.

To this end, first write a tensor product smooth as

$$f(\mathbf{x}) = \sum_{j=1}^{J} b_j(\mathbf{x})\beta_j$$

with associated wiggliness measure $J(f) = \sum_{i=1}^{d} \lambda_i \boldsymbol{\beta}^{\mathrm{T}} \tilde{\mathbf{S}}_i \boldsymbol{\beta}$, where each $\tilde{\mathbf{S}}_i$ is a positive semidefinite matrix of coefficients (only semidefinite because most penalties treat some space of functions as having zero wiggliness). Given $(y_i, \mathbf{x}_i)$ data, it is straightforward to produce a model matrix $\mathbf{X}^f$, where $X^f_{ij} = b_j(\mathbf{x}_i)$, so that $\mathbf{X}^f\boldsymbol{\beta}$ is a vector of $f(\mathbf{x}_i)$ values.

The mixed model approach to estimating f starts from the premise that, by stating that f is smooth, we really believe that it is more probable that f is smooth than that f is wiggly. This can be formalized by specifying a prior for the wiggliness of the model, which is, say, $\propto \exp(-\sum_i \lambda_i \boldsymbol{\beta}^{\mathrm{T}} \tilde{\mathbf{S}}_i \boldsymbol{\beta}/2)$. Such a prior implies an improper Gaussian prior for $\boldsymbol{\beta}$ itself (Silverman, 1985). This improper distribution for $\boldsymbol{\beta}$ does not fit easily into standard linear mixed modeling approaches (e.g., Pinheiro and Bates, 2000). Some reparameterization is therefore needed, so that the new parameters divide into a set with a proper distribution, to be treated as random effects, and a set (of size M_T) with an improper uniform distribution, which can be treated as fixed effects.

To achieve this, first note that the degree of rank deficiency of $\sum_i \lambda_i \tilde{\mathbf{S}}_i$ is readily shown to be M_T, the product of the dimensions of the null spaces of the marginal penalty matrices $\tilde{\mathbf{S}}_i$ (provided that $\lambda_i > 0\ \forall i$). Reparameterization is based on the eigendecomposition

$$\sum_{i=1}^{d} \tilde{\mathbf{S}}_i = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathrm{T}},$$

where $\mathbf{U}$ is an orthogonal matrix of eigenvectors and $\mathbf{D}$ is a diagonal matrix of eigenvalues, with M_T zero elements at the end of the leading diagonal. Notice that there are no λ_i parameters in the sum that is decomposed: this is reasonable since the null space of the penalty does not depend on these parameters (however, given finite precision arithmetic it might be necessary to scale the $\tilde{\mathbf{S}}_i$ matrices in some cases).

Partitioning the eigenvector matrix so that $\mathbf{U} \equiv [\mathbf{U}_R : \mathbf{U}_F]$ where $\mathbf{U}_F$ has M_T columns, it is necessary to define $\mathbf{X}_F \equiv \mathbf{X}^f\mathbf{U}_F$, $\mathbf{Z} \equiv \mathbf{X}^f\mathbf{U}_R$, and $\boldsymbol{\mathcal{S}}_i = \mathbf{U}_R^{\mathrm{T}}\tilde{\mathbf{S}}_i\mathbf{U}_R$. A mixed model representation of the tensor product term (i.e., the linear predictor and random effects distribution) is

$$\mathbf{X}_F\boldsymbol{\beta}_F + \mathbf{Z}\mathbf{b}, \quad \mathbf{b} \sim N\left(\mathbf{0}, \left(\sum \lambda_i \boldsymbol{\mathcal{S}}_i\right)^{-1}\right),$$

where the λ_i and $\boldsymbol{\beta}_F$ parameters have to be estimated. Clearly the covariance matrix structure is not completely standard, but neither is it difficult to implement. For example, a new `pdMat` class implementing this covariance matrix structure can readily be written for use with the `nlme` software of Pinheiro and Bates (2000). Given such a class, incorporation of one or more tensor product terms into a linear mixed model is straightforward.

3.1 *Generalization and Confidence Intervals*

The discussion so far has focused on additive mixed models, and general methods for setting these up in a manner allowing estimation using standard software such as the `nlme` library of Pinheiro and Bates (2000). Estimation in the generalized case can proceed in a completely straightforward manner using the approximate penalized quasi-likelihood (PQL) methods of, for example, Breslow and Clayton (1993). Venables and Ripley (2002) provide a suitable function `glmmPQL` based on iterative calls to the mixed modeling function `lme` from the `nlme` library.

The remaining issue is the calculation of confidence intervals. In most applications of GAMMs these would be required primarily for the smooth terms and the fixed effects. If this is the case then, following Silverman (1985), a Bayesian posterior covariance matrix for the coefficients of these terms can be obtained. Conditioning on the parameter estimates for the random effects, it is first necessary to calculate the covariance matrix for the response data (or pseudodata in the PQL case) implied by the estimated random effects structure *excluding* the smooth terms; suppose this is $\mathbf{V}$. Then if $\boldsymbol{\theta}$ is the vector of all the fixed parameters plus the coefficient of the smooths,

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$\mathbf{X}$ is the model matrix corresponding to these terms, and $\mathbf{S}_i$ is the ith penalty matrix (padded with zeros if necessary so that $\boldsymbol{\theta}^{\mathrm{T}}\mathbf{S}_i\boldsymbol{\theta}$ is the correct penalty) then

$$\boldsymbol{\theta}\,|\,\mathbf{y} \sim N\left(\hat{\boldsymbol{\theta}}, \left(\mathbf{X}^{\mathrm{T}}\mathbf{V}^{-1}\mathbf{X} + \sum \lambda_i \mathbf{S}_i\right)^{-1}\right),$$

where $\hat{\boldsymbol{\theta}}$ is the vector of estimates or predictions of the elements of $\boldsymbol{\theta}$. This is essentially the approach taken in Lin and Zhang (1999), and allows the required intervals to be obtained. The only quantity not readily available from standard software is the estimate $\mathbf{V}$, but with some effort it is possible to extract it, at least from `lme` fits. As usual the degrees of freedom per element of $\boldsymbol{\theta}$ can be estimated from the leading diagonal of $(\mathbf{X}^{\mathrm{T}}\mathbf{V}^{-1}\mathbf{X} + \sum \lambda_i \mathbf{S}_i)^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{V}^{-1}\mathbf{X}$.

4. Comparison with Existing Techniques

There is a substantial practical difference between using the approach suggested in Section 2 and a simpler tensor product approach employing a single penalty matrix such as $\mathbf{S}_\pi = \mathbf{S}_w \otimes \mathbf{S}_x \otimes \mathbf{S}_z$ (e.g., Fahrmeier and Lang, 2001). The problem with such single penalties is their degree of rank deficiency. For example, a smooth of three variables constructed from three cubic spline bases, each of rank 5, would have 125 parameters and a penalty of rank 27. Hence the effective degrees of freedom of the term would have to lie between 98 and 125, rendering the penalization effectively useless. In contrast, using the same marginal bases and the approach advocated here, the degrees of freedom of the smooth would lie between 8 and 125, a much more useful range for practical work. Alternatively one could employ a higher-rank single penalty in association with a tensor product basis, but in that case the resulting smooth is no longer invariant to linear rescaling of the arguments of the smooth. All proposals for tensor product smoothing with a single penalty are unsatisfactory for this reason; it is possible to have a useful tuneable smoothness range or scale invariance, but not both, and for practical purposes both are needed.

4.1 *Tensor Products of P-Splines*

Eilers and Marx (2003) proposed constructing smooths of two variables from tensor products of P-splines, with separate difference penalties applied to each covariate direction, and hence two smoothing parameters to estimate. This suggestion is a special case of the method given in Section 2, but without the reparameterization (or integration) step in the construction of the penalty. A small simulation comparing the Eilers and Marx method and the method given in Section 2 was performed. Four hundred data were simulated from f_1 from Section 4.2 with additive independent and identically distributed (i.i.d.) Gaussian noise ($\sigma = 1$). Two hundred replicate data sets were produced and each reconstruction of f_1 was attempted using both alternative methods. Each employed P-splines with a cubic basis and second-order penalties as marginal penalties. Smoothing parameters were estimated by generalized cross-validation (GCV). Mean square error (MSE) performance was on average 8.5% better using the Section 2 smooths, and these outperformed the Eilers and Marx smooths in 83% of replicates. Similar modest improvements were obtained at other noise levels, and are presumably attributable to the care that has been taken to preserve the penalties' relationship to function shape when moving from the marginal smooths to the tensor product smooth in Section 2.

4.2 *Smoothing Spline ANOVA*

Smoothing spline ANOVA, for which Gu (2002) is an excellent guide, provides a self-contained framework for modeling with smooth functions, including an approach to mixed modeling. The tensor product smooths used in this framework have the scale invariance properties of the smooths introduced in Section 2, although a different construction is used to achieve this. Until recently these methods were rather computationally costly, but Kim and Gu (2004) have substantially improved this situation using low-rank approximations, so a small simulation study was undertaken to compare SS-ANOVA to the methods proposed in Section 2, in a penalized regression context. Data were simulated from four models of the general form

$$y_i = f_j(x_i, z_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

where the ϵ_i were independent, the covariates x and z were independent uniform random deviates on (0, 1), and the functions f_j were

$$f_1(x,z) = 10\pi\sigma_x\sigma_z\left(1.2e^{-(x-0.2)^2/\sigma_x^2-(z-0.3)^2/\sigma_z^2} + 0.8e^{-(x-0.7)^2/\sigma_x^2-(z-0.8)^2/\sigma_z^2}\right),$$

$$f_2(x,z) = 2\sin(\pi * x) + \exp(2z),$$

$$f_3(x,z) = 1.9(1.45 + \exp(x)\sin(13(x-0.6)^2))\exp(-z)\sin(7z)$$

and $f_4 = (f_1 + f_2)/2$ ($\sigma_x = 0.3$ and $\sigma_z = 0.4$). Two hundred replicate simulations were made using each model for each combination of $\sigma = 2$ and $\sigma = 0.5$ with sample sizes 200 and 1000. For each replicate, four reconstructions of f_j were produced and their mean square prediction error at the covariate values was assessed. The four approaches were (i) fit an SS-ANOVA model, by Kim and Gu's method as implemented in routine `ssanova1` of Gu's `R` package `gss`, (ii) fit a single Section 2 smooth, (iii) fit a similar ANOVA decomposition model to that used in (i), but based on Section 2 smooths, as described in Section 2.1, and (iv) choose whichever of (ii) and (iii) has the lowest GCV score. All alternatives used GCV to estimate smoothing parameters, and models (ii) and (iii) were estimated by the method of Wood (2004) as implemented in `R` package `mgcv`. All models used a basis dimension of 49.

The results are shown in Figure 2. Method (iv) gave better average MSE performance than method (i) in all cases, and required between 1/4 (n = 200) and 1/12 (n = 1000) of the computation time. In fact method (ii) was only worse than method (i) in terms of MSE for the strictly additive f_2, but required between 1/10 and 1/30 of the computation time.

A smaller three-covariate simulation was also undertaken. Five hundred triples x_i, z_i, v_i were generated from independent uniform distributions on (0, 1) and response data were generated from

$$y_i = 1.5e^{-(x_i-0.2)^2/5-(z_i-0.5)^2/3-(v_i-0.9)^2/4} + 0.5e^{-(x_i-0.3)^2/4-(z_i-0.7)^2/2-(v_i-0.4)^2/6} + e^{-(x_i-0.1)^2/5-(z_i-0.3)^2/5-(v_i-0.7)^2/4} + \epsilon_i,$$

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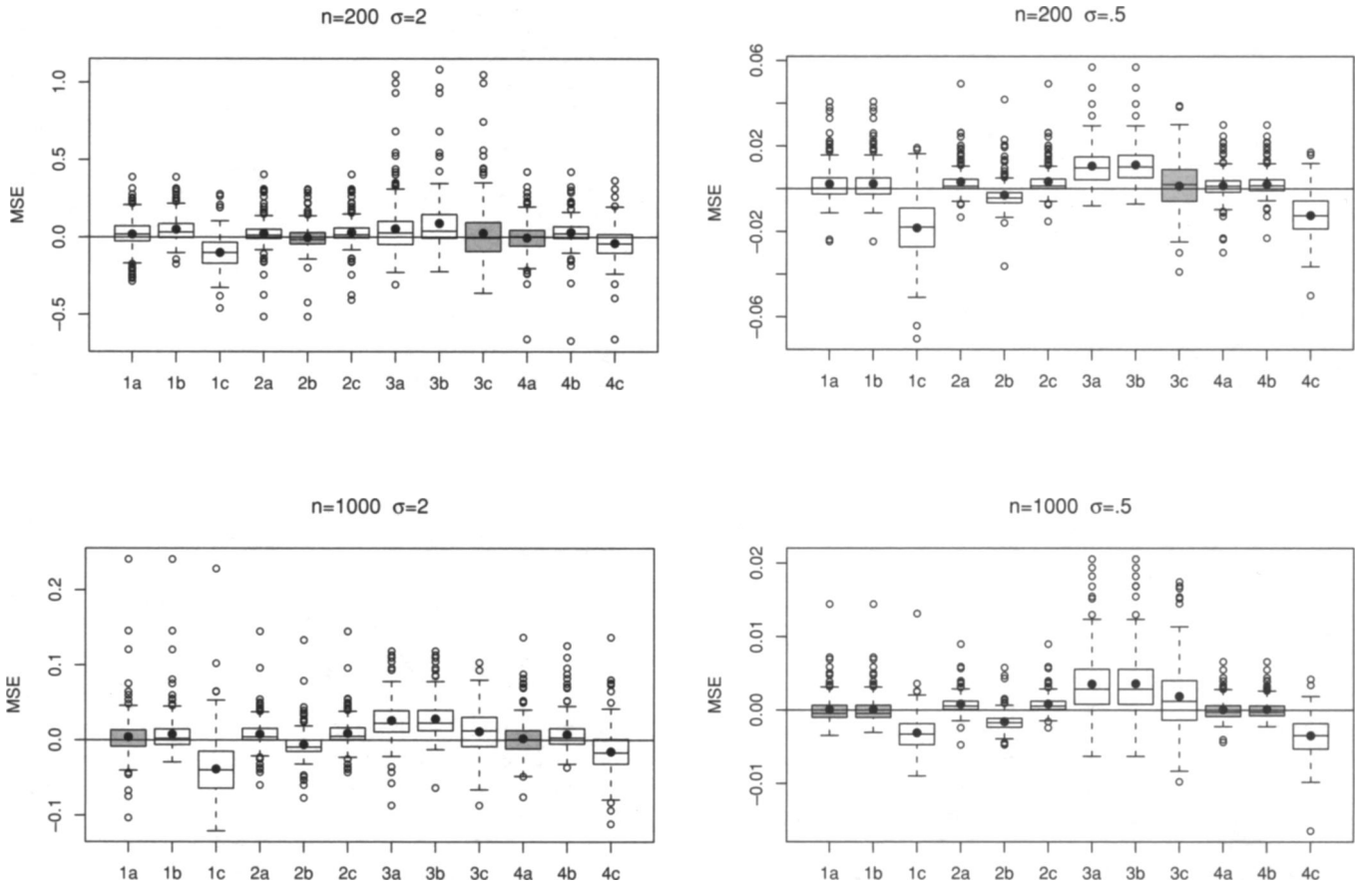

Figure 2. Comparisons of mean square error (MSE) performance for the tensor product smooths as given in Section 2 and the low-rank SS-ANOVA method of Kim and Gu (2004) for smooth functions of two covariates. Each boxplot shows the SS-ANOVA MSE minus the MSE of the alternative method based on the smooths in Section 2 over 200 replicates. The numbers in the x-axes labels refer to the index of the true function from which the data were simulated. The letters in the labels are as follows: (a) is where whichever of a single smooth and an SS-ANOVA style decomposition of a smooth had the lower GCV score was selected as the best model; (b) is for a single Section 2 style smooth; (c) is for a smooth represented using an SS-ANOVA-type decomposition, but using the Section 2 smooths to represent this. The four panels relate to different combinations of sample size and noise level. All values above the zero line indicate that the Section-2-based method had better MSE than the Kim and Gu SS-ANOVA method. Gray-shaded boxplots indicate comparisons where the mean difference in MSE was not significantly different from zero at the 5% level. The type (a) smooths, based on the methods given in Section 2, consistently outperform the Kim and Gu (2004) SS-ANOVA method, while using less computer time to estimate.

where the ϵ_i are i.i.d. normal random deviates with $\sigma = 0.02$. One hundred replicates of this simulation were performed. Three types of models were fitted to each replicate: (a) a tensor product smooth of x_i, z_i, and v_i; (b) an SS-ANOVA-type model with all seven terms represented using the tensor product smooths given in Section 2; and (c) an SS-ANOVA model (of similar structure to that used in [b]) estimated using the method of Kim and Gu (2004). Kim and Gu recommend that, to avoid overfitting, each model degree of freedom should be treated as if it were 1.4 degrees of freedom when calculating GCV scores, so each model was estimated both by conventional GCV and this modified version (for the two covariate simulations the modified GCV score provided no improvement, possibly because rather fewer smoothing parameters are estimated in that case). For this example all models had a basis dimension of 64.

The results are summarized in Figure 3. In this case method (c) took 190 times as long as method (b) and 550 times as long as method (a). Method (b) had the best MSE performance followed by method (c). With so many smoothing parameters in the ANOVA-type models (12 for [b] and 19 for [c]), the nature of the penalty is effectively being estimated, as well as the degree of smoothness, and it may be this degree of flexibility that allows better MSE performance.

Experimentation with a four-covariate smooth produced similar results: the 65-smoothing-parameter, full SS-ANOVA becomes even more costly in this case, and again, better MSE performance seems to be obtainable more cheaply using the smooths as given in Section 2.

To conclude, the methods presented in this article provide a useful complementary approach to the smoothing spline ANOVA framework described in Gu (2002) and may

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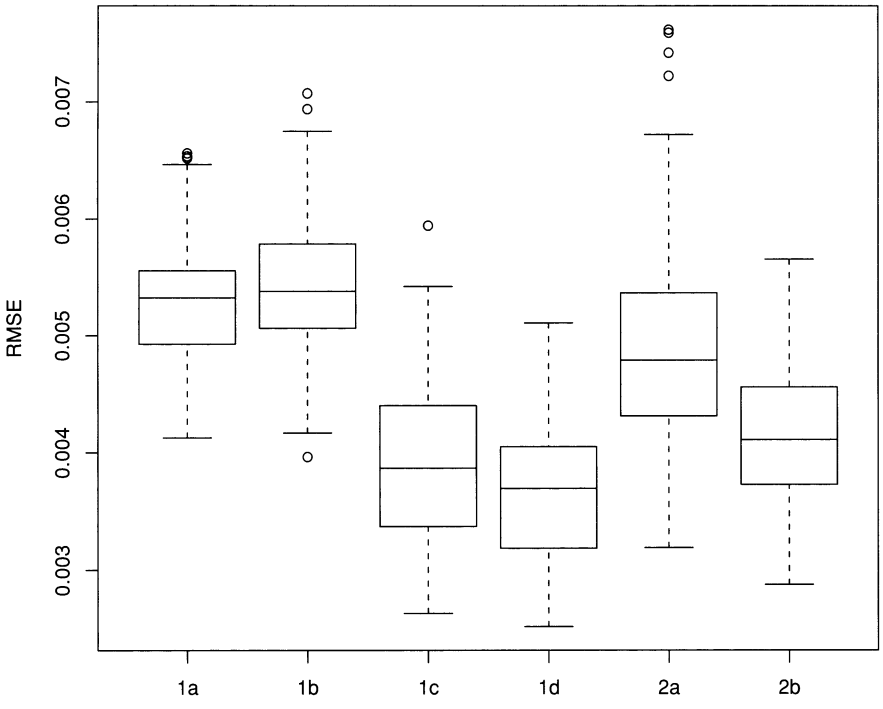

Figure 3. Comparisons of root mean square error (RMSE) performance for the tensor product smooths as given in Section 2 and the low-rank SS-ANOVA method of Kim and Gu (2004) for a smooth function of three covariates. Each boxplot shows the RMSE for one alternative model. Smoothing parameters were estimated by GCV or vGCV in which each model degree of freedom counts as 1.4 degrees of freedom. 1a and 1b are for single Section 2 smooths estimated by GCV and vGCV, respectively; 1c and 1d are for SS-ANOVA style decompositions represented using Section 2 smooths and estimated by GCV and vGCV, respectively; 2a and 2b are for Kim and Gu SS-ANOVA models estimated by GCV and vGCV, respectively. 1c consistently gave lower GCV scores than 1a, while 1d consistently gave lower vGCV scores than 1b.

offer the following advantages: (i) in a penalized regression context, the simulation study suggests that the new methods can compare favorably with SS-ANOVA in both computational speed and MSE performance; (ii) because the bases of the smooths do not depend on the smoothing parameters, the new smooths are readily incorporated into existing frameworks for linear and generalized linear mixed modeling, and indeed nonlinear mixed effects modeling; this in turn gives ready access to a rich variety of modeling tools including all the inferential machinery of mixed models, and the reliable and extensive software libraries for mixed modeling; (iii) the new tensor product wiggliness penalties are relatively interpretable in terms of function shape, provided that the penalties associated with the marginal smooths are interpretable, whereas the penalties associated with full SS-ANOVA models require quite a high degree of sophistication to interpret; (iv) the interpretation of the distribution of the model coefficients treated as random effects is straightforward: a negative exponential distribution is assumed for function "wiggliness"; and (v) the new methods make it very easy to build smooths of several variables from smooths designed for particular modeling tasks, for example, for some practical applications smooths should be very "flat" away from high densities of supporting data, and the new methods can easily accommodate such "designer smooths."

5. Examples

This section illustrates the utility of the methods using two, very brief, examples, one simulated and the other real. The first example simulates a repeated measures situation, such as might occur in one arm of a blood pressure trial. y_{ij} is the jth measurement on the ith subject, and is governed by

$$y_{ij} = \mu_{ij} + \epsilon_{i,j}, \text{ where } \log(\mu_{ij}) = f_1(t_{ij}, w_i) + f_2(z_i) + b_i t_{ij},$$

where the b_i are i.i.d. $N(0, \sigma_b^2)$ and $\epsilon_{i,j} = \phi\epsilon_{i,j-1} + e_{ij}$, where the e_{ij} are i.i.d. $N(0, \sigma^2)$. f_1 and f_2 are smooth functions of the covariates t, w, and z. Twenty-four measurements were simulated for each of 50 subjects from this model, with $\sigma_b = 10$, $\sigma = 5$, and $\phi = 0.6$; f_1 and f_2 are as shown in Figure 4.

f_1 and f_2 were estimated from these simulated data by fitting the model with f_2 represented as a cubic regression spline and f_1 represented using either a tensor product constructed from cubic regression splines or a thin plate regression spline (Wood, 2003). Figure 4 shows various reconstructions for a typical replicate simulation. Notice how the tensor product smooth outperforms the isotropic smooth as a result of the rather different degree of wiggliness of the true f_1 with respect to a unit change on the different axes; the best that the isotropic smoother can do is to model f_1 with a plane. Similarly, neglecting the autocorrelation and random subject effects leads to overfitting.

This example serves to highlight the utility of the scale invariance of the tensor product smooths proposed in this article, as well as the usefulness of ready access to facilities for properly modeling random effects and residual autocorrelation. The low rank of the model is also useful here: a full spline approach would have required estimation of 1200

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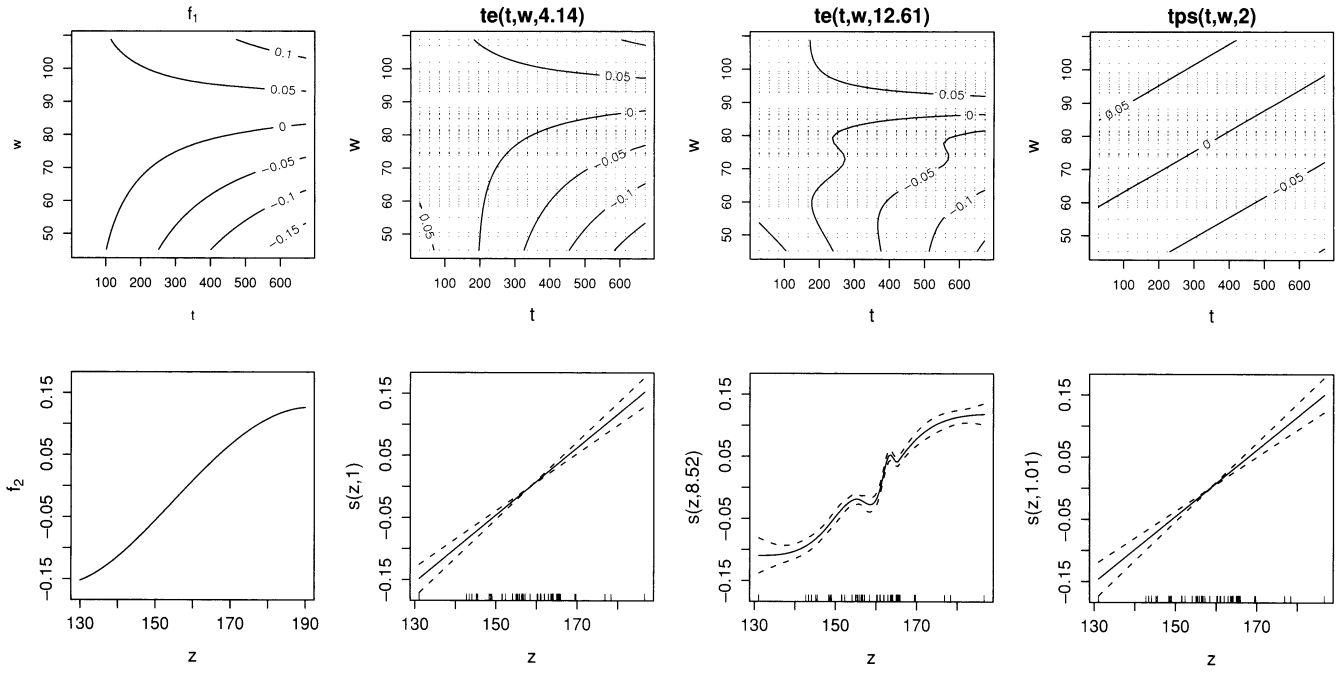

Figure 4. The true and estimated component functions of the simulated repeated measures GAMM example. The top row relates to f_1 and the bottom row to f_2. The columns, from left to right, show: the true functions used in simulation; the component functions of a GAMM, estimated by PQL, with f_1 represented as a tensor product term, and with a random time effect for subject and AR1 residual correlation; the same as the previous column, but assuming no random subject effect and i.i.d. errors; finally, the same as column 2, but with an isotropic smooth term for f_1. In all cases the figure in the response axis label gives the effective degrees of freedom of the plotted smooth-term estimate. For each reconstruction the basis dimension was 49 for f_2 and 10 for f_1. Covariate values are shown as points on the f_1 plots, and rug plots on the f_2 plots. Ninety-five percent credible regions are shown on the $\hat{f}_2$ plots.

model coefficients, where computational cost is cubic in the number of coefficients.

5.1 *Fish Egg Survey Example*

Fish stock assessments are sometimes undertaken by surveying the eggs of a particular species in order to work out egg abundance, from which total mass of the spawning stock of fish can be inferred. One such survey was undertaken in 1990 in the Bristol Channel (southwest England), targeting Sole eggs. Five research cruises were undertaken spread over the Sole spawning season. On each cruise, eggs were sampled by vertical net hauls at each of a roughly even grid of sampling stations spread over the spawning area. Each net haul would potentially sample eggs from each of the four identifiable age classes.

The primary interest in such surveys is to estimate the rate of egg spawning over the survey area, which can be related (approximately) to the observed egg age class densities by the following: if y_{ij} is the observed density (per m^2 sea surface) of stage j eggs at sampling station i, then $E(y_{ij} \mid b_i) = \mu_{ij}$, where

$$\mu_{ij} = \Delta_{ij} R(\mathrm{lo}_i, \mathrm{la}_i, t_i) e^{-\delta(t_i) a_{ij}} b_i. \tag{3}$$

R is the spawning rate at the location (lo_i, la_i) and time t_i of the ith sample and δ is the per capita mortality rate, which is assumed to depend only on time. Δ_{ij} is the duration of the jth age class at the ith sampling station, while a_{ij} is the mid-point age of the class. The $\log(b_i)$ are i.i.d. $N(0, \sigma_b^2)$ random effects for sampling station. R and δ are assumed to be smooth functions of their arguments. The variance of $y_{ij} \mid b_i$ is assumed proportional to μ_{ij}. A full explanation of the mechanisms underpinning this model and of the data can be found in Horwood and Greer Walker (1990), Horwood (1993), and Dixon (2003).

By taking logs of (3) it can be linearized so that

$$\log(\mu_{ij}) = \log(\Delta_{ij}) + f_1(\mathrm{lo}_i, \mathrm{la}_i, t_i) - \delta(t_i) a_{ij} + \log(b_i),$$

where $f_1 = \log(R)$. The $\log(\Delta_{ij})$ term can be treated as a known offset. Clearly the model now has the general structure of a GAMM, with the slightly unusual feature that the smooth function $\delta(t)$ is multiplied by the known covariate $-a_{ij}$, so that the model is a mixed version of a variable coefficient model (Hastie and Tibshirani, 1993).

For estimation purposes, f_1 was represented using the tensor product smooth constructed from three rank 5 cubic regression spline marginal bases, while δ was represented using a rank 10 cubic regression spline. Penalty construction and representation as mixed model components followed the methods described above. f_1, δ, and σ_b^2 were then estimated by fitting the model by PQL. Figure 5 shows the contour plots of $\hat{R} = \exp(\hat{f}_1)$ over the Bristol Channel at various times, throughout the spawning season. The maps themselves are useful for management purposes, but once the model is estimated then upper and lower credible limits are also easily

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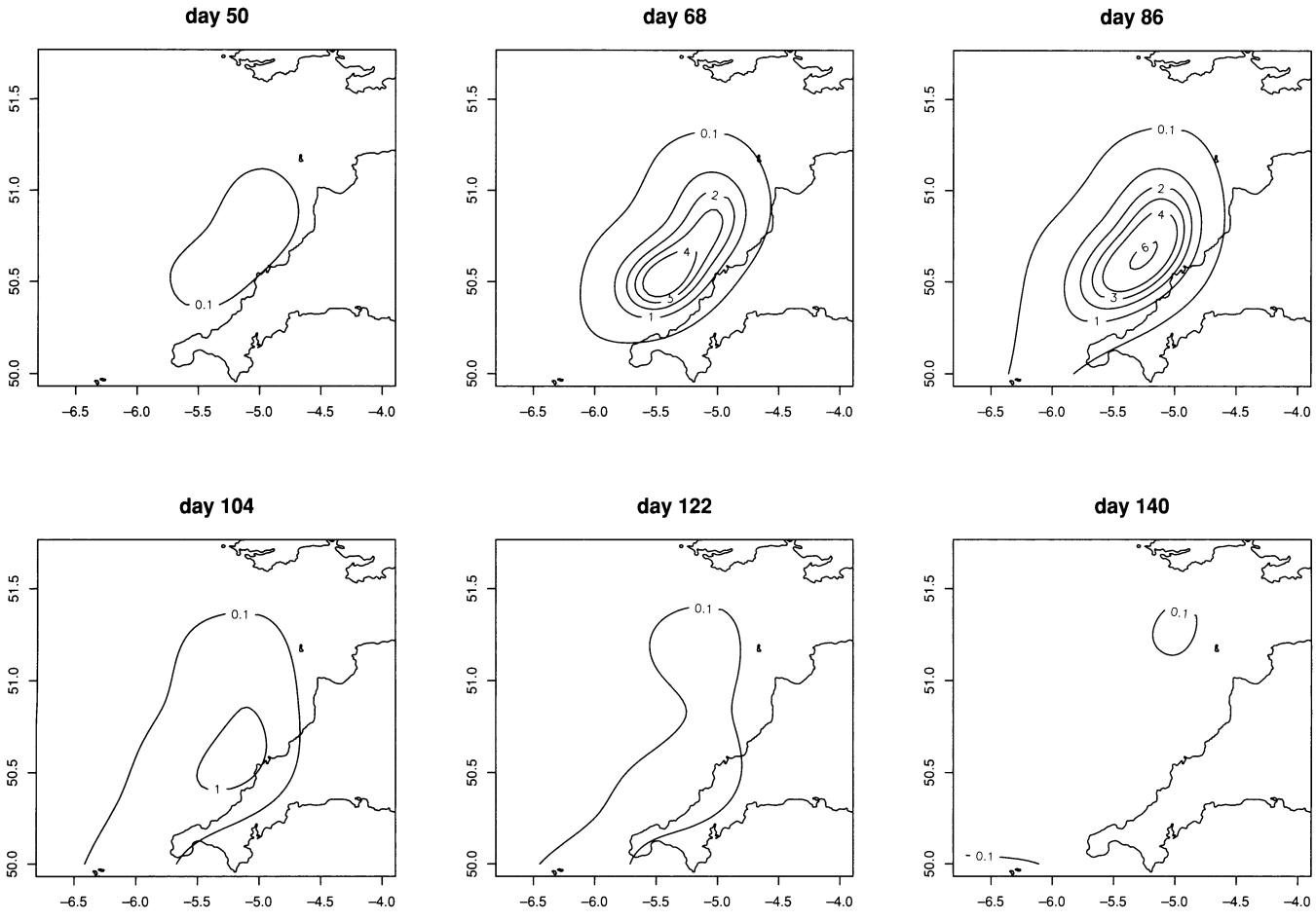

Figure 5. Estimated spawning rate (eggs d^{-1} m^{-2}) of Sole eggs over the Bristol Channel in the 1990 spawning season, from the model estimated in Section 5.1. The days quoted are Julian days (i.e., day 1 is January 1, 1990). Notice the tendency for the distribution to become more diffuse as spawning progresses. Maps like this can provide valuable input when making decisions about management by restrictions on the areas where fishing is allowed.

produced for each map, as well as credible intervals for the total egg production over the year.

6. Conclusions

The main innovation reported in this article is a general method for producing low-rank, scale-invariant tensor product smooths for inclusion into GAMMs and GAMs, which have a practically useful smoothness range when smoothness is to be estimated as part of model fitting, and which can be constructed from a wide variety of "marginal" smooths. The importance of scale invariance is well illustrated in the first row of Figure 4, where an example of the suggested smooths is compared with an isotropic smooth: if the covariates of a smooth are not on the same scale, then assuming isotropy can lead to very poor results, which the proposed method overcomes by using a separate penalty for each covariate direction. Although the representation of smooths as mixed model components is well established, the reparameterization that allows this to be done for the general tensor product smooths as given in Section 2 is novel. The consequent ability to include tensor product smooths directly into standard mixed models is practically useful, both as an extension to what mixed models can do and because of the ready access that it gives to mixed modeling theory for tasks such as smoothing parameter confidence interval calculation. Without the methods presented in this article, the task of estimating the models presented in Section 5 would be rather formidable, given currently available software.

The methods given in Section 2 can be viewed as a generalization of the proposal of Eilers and Marx (2003) to any number of covariates, any marginal smoothing bases and quadratic penalties, and mixed model settings, while offering improvements in the performance of such smooths by virtue of the novel approach to penalty construction. As discussed in Section 4.2, the methods complement existing SS-ANOVA methods by providing a means of representing tensor product smooths and ANOVA decompositions of functions, which is easily extended (e.g., to variable coefficient models and nonlinear mixed models) and easily incorporated into existing mixed modeling methodology, while allowing considerable freedom to design smooths for particular purposes. The simulation study suggests that, when compared to existing SS-ANOVA models, the new methods can perform favorably in terms of computation time and MSE performance. These

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Table 1

Number of effective degrees of freedom required to reconstruct the function $f(\mathbf{x}) = \exp(-||\mathbf{x}||^2/2)$ from n data on $[-2, 2]^d$, where $d = \dim(\mathbf{x})$, using thin plate splines, $\hat{f}$, operating to reconstruction accuracy ϵ

	$n = 300$		$n = 500$	
d	$\epsilon = 0.05$	$\epsilon = 0.02$	$\epsilon = 0.05$	$\epsilon = 0.02$
1	5.2	5.9	5.2	5.9
2	19.4	26.1	19.4	26.9
3	74.6	111.3	79.4	130.6

Note: The data were noiseless observations of $f(\mathbf{x})$ at n values of $\mathbf{x}$ simulated from a uniform distribution on $[-2, 2]^d$. The thin plate spline smoothing parameters were chosen so that $\epsilon = \max_i |f(\mathbf{x}_i) - \hat{f}(\mathbf{x}_i)|$ was either 0.05 or 0.02, that is, to give a specified degree of approximation error (the measure is chosen for artifact-free comparability between different values of d). All figures are averages over 20 replicates. Notice that the number of degrees of freedom required to estimate f to given accuracy grows approximately exponentially with d: this is fundamental to nonparametric function estimation, rather than being a special feature of tensor product smoothing.

simulation results should, of course, be treated with some circumspection: it is quite possible that with very uneven configurations of covariates the SS-ANOVA method's better adaptation to covariate configuration might lead to a reversal of the apparent MSE advantage, at least for smooths of several covariates.

From the tensor product construction as given in Section 2, it seems natural that the size of the tensor product basis should increase exponentially with the number of covariates of a smooth. Superficially, this "curse of dimensionality" appears to be an alarming deficiency of the method, but in fact it does no more than reflect a fundamental property of nonparametric function estimation. The number of degrees of freedom *required* to nonparametrically estimate functions of different numbers of covariates, to a given accuracy, increases exponentially with the number of covariates. Table 1 provides an easily replicated illustration of this phenomenon using thin plate spline smoothers, while Hastie and Tibshirani (1990, Chapter 4) provide a good discussion of the "curse of dimensionality" in the context of nonparametric function estimation.

The computational cost of penalized regression smoothers of several covariates is generally of cubic order in the basis dimension, so the low-rank smooths presented here are generally very computationally efficient, relative to full smoothing splines, say, where the basis dimension is the number of data. The price for this efficiency is that the basis dimension has to be chosen. The exact choice is not generally critical: it is the smoothing parameters that determine the effective degrees of freedom of the smooth with the basis dimension simply setting an upper limit, and what matters is that this upper limit is not overly restrictive. Unfortunately what counts as "overly restrictive" depends on the context, and in particular the complexity of the unknown truth that it is hoped to estimate. In the case of some spline-based models Gu and Kim (2002) derive the result that basis size should be $kn^{2/9}$, where n is sample size, but unfortunately k is an unknown constant, which really depends on the unknown truth being estimated. As with many aspects of modeling a choice has to be made based on knowledge of the system being modeled, and the adequacy of the choice reviewed in the light of model fit.

The methods described here are implemented in package `mgcv` for `R` (R Core Development Team, 2003).

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References

Ball, R. D. (2003). lmeSplines. *R News* **3**(3), 24–28.

Borchers, D. L., Buckland, S. T., Priede, I. G., and Ahmadi, S. (1997). Improving the precision of the daily egg production method using generalized additive models. *Canadian Journal of Fisheries and Aquatic Science* **54,** 2727–2742.

Breslow, N. E. and Clayton, D. G. (1993). Approximate inference in generalized linear mixed models. *Journal of the American Statistical Association* **88,** 9–25.

Dixon, C. E. (2003). Multi-dimensional modelling of physiologically and temporally structured populations. Ph.D. Thesis, University of St Andrews, U.K.

Eilers, P. H. C. and Marx, B. D. (1996). Flexible smoothing with B-splines and penalties. *Statistical Science* **11,** 89–121.

Eilers, P. H. C. and Marx, B. D. (2003). Multivariate calibration with temperature interaction using two-dimensional penalized signal regression. *Chemometrics and Intelligent Laboratory Systems* **66,** 159–174.

Fahrmeir, L. and Lang, S. (2001). Bayesian inference for generalized additive mixed models based on Markov random field priors. *Applied Statistics* **50,** 201–220.

Fahrmeir, L., Kneib, T., and Lang, S. (2004) Penalized structured additive regression for space-time data: A Bayesian perspective. *Statistica Sinica* **14,** 731–761.

Gu, C. (2002). *Smoothing Spline ANOVA Models.* New York: Springer-Verlag.

Gu, C. and Kim, Y. J. (2002). Penalized likelihood regression: General approximation and efficient approximation. *Canadian Journal of Statistics* **34,** 619–628.

Hastie, T. J. and Tibshirani, R. J. (1990). *Generalized Additive Models.* London: Chapman & Hall.

Hastie, T. J. and Tibshirani, R. J. (1993). Varying coefficient models. *Journal of the Royal Statistical Society, Series B* **55,** 757–796.

Horwood, J. (1993). The Bristol Channel sole (*Solea solea (L.)*): A fisheries case study. *Advances in Marine Biology* **29,** 215–367.

Horwood, J. and Greer Walker, M. (1990). Determinacy of fecundity in sole (*Solea solea*) from the Bristol Channel.

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Journal of the Marine Biological Association of the UK **70,** 803–813.

Kammann, E. E. and Wand, M. P. (2003). Geoadditive models. *Applied Statistics* **52,** 1–18.

Kim, Y. J. and Gu, C. (2004). Smoothing spline Gaussian regression: More scalable computation via efficient approximation. *Journal of the Royal Statistical Society, Series B* **66,** 337–356.

Lang, S. and Bresger, D. (2004). Bayesian P-splines. *Journal of Computational and Graphical Statistics* **13,** 183–212.

Lin, X. and Zhang, D. (1999). Inference in generalized additive mixed models using smoothing splines. *Journal of the Royal Statistical Society, Series B* **61,** 381–400.

Pinheiro, J. C. and Bates, D. M. (2000). *Mixed-Effects Models in S and S-PLUS.* New York: Springer-Verlag.

R Development Core Team. (2003). *R: A Language and Environment for Statistical Computing.* Vienna: R Foundation for Statistical Computing. Available at `http://www.R-project.org`

Ruppert, D., Wand, M. P., and Carroll, R. J. (2003). *Semiparametric Regression.* Cambridge, U.K.: Cambridge University Press.

Silverman, B. W. (1985). Some aspects of the spline smoothing approach to nonparametric regression curve fitting. *Journal of the Royal Statistical Society, Series B* **47,** 1–52.

Venables, W. N. and Ripley, B. D. (2002). *Modern Applied Statistics in S*, 4th edition. New York: Springer-Verlag.

Verbyla, A. P., Cullis, B. R., Kenward, M. G., and Welham, S. J. (1999). The analysis of designed experiments and longitudinal data by using smoothing splines. *Applied Statistics* **48,** 269–313.

Wahba, G. (1990). *Spline Models for Observational Data.* Philadelphia: SIAM.

Wang, Y. (1998). Mixed effects smoothing spline analysis of variance. *Journal of the Royal Statistical Society, Series B* **60,** 159–174.

Wood, S. N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society, Series B* **65,** 95–114.

Wood, S. N. (2004). Stable and efficient multiple smoothing parameter estimation for generalized additive models. *Journal of the American Statistical Association* **99,** 637–686.

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